

One Stage, One Act: Independent Conditioning Channels Do Not Add on a Frozen Language Model

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Abstract

This note expands on papers 7 and 8 [1, 2]. Paper 7 closed additive weight-space composition of sensor bricks on a frozen language model; paper 8 showed selection engages composition exactly. This note runs the program’s two proven conditioning mechanisms *simultaneously* for the first time — a hypernetwork sensor block (a weight-space channel) and a model-to-model latent packet (an activation-space channel) on one frozen 35B host — and then merges two packets in the activation channel alone. The results extend the negative to its general form: two packets sharing one write budget destroy each other (joint heed 1/16); doubling the budget yields dominance of the stronger packet, never coexistence. Across channels the outcome is governed by *behavioral dominance*, not stream interference: the weakest block coexists with the packet (retention 0.82/0.92), while strong blocks — with or without affect — capture the response and halve or worse the packet’s effect (retention 0.50 and 0.30). Scaling a block’s write strength β recovers the packet monotonically on both blocks tested, so dominance is *dosed* — but β does not equalize blocks, so it is not *fungible*. The constructive half survives intact: one act at a time composes exactly, an arbiter can choose it (paper 8), and this note adds that the arbiter has a volume knob. What dies, with proof, is superposition itself — in weight space, in activation space, and across the two.

1 Why this note exists

The program has two mechanisms that provably condition a frozen model. The **weight channel**: a hypernetwork reads sensor features and writes a LoRA delta onto the host’s attention projections (papers 1–3, 7, 8). The **activation channel**: a small frozen model emits a 256-dimensional latent packet, injected into the host’s residual stream at three layers, carrying information the host never sees as text; its gate passed at the 2B→35B scale with true joint 0.875 on held-out phrasings under topic-partner shuffled controls (program artifacts; operating dose 0.20). Every composition result to date lives inside one channel — and inside the weight channel, addition is exhaustively dead while selection is exact. Three registered questions follow: do the two channels coexist on one host; do two packets merge inside the activation channel; and if channels conflict, what governs the outcome. All three ran preregistered on 2026-08-05, evaluation-only, from checkpoints that already existed. Because bf16 MoE greedy decode is not bit-reproducible across processes, every comparison below is in-band — solo baselines re-decoded in the same process as the dual cells — and the recorded anchors are reported beside them (they matched to the item in the first session and within the documented two-item band in the rest).

2 Experiment 1: one block and one packet

Three production blocks from the 35B r2 column were paired with the packet at its recorded operating dose, each side scored by its own registered instrument with the other channel active (Table 1).

block	relay joint, block on	rel. ret.	block eval, packet off	block eval, packet on	blk. ret.
temporal	23/32 (0.719 $\pm$ 0.149)	0.821	12/15 (0.800 $\pm$ 0.191)	11/15 (0.733 $\pm$ 0.205)	0.917
self	14/32 (0.438 $\pm$ 0.163)	0.500	18/18 (1.000 $\pm$ 0.088)	17/18 (0.944 $\pm$ 0.124)	0.944
geo	9/32 (0.281 $\pm$ 0.149)	0.300	18/18 (1.000 $\pm$ 0.088)	17/18 (0.944 $\pm$ 0.124)	0.944

Table 1: Each channel with the other active. Relay solo (in-band, block off): 28/32 (0.875 $\pm$ 0.115) in the first session, 30/32 (0.938 $\pm$ 0.092) in the geo session — the documented cross-process band. Retention $\geq$ 0.80 was the registered survival bar.

The verdict splits, and the split is the finding. **temporal** — the weakest block on its own evaluation — coexists: both channels clear the bar. **self** and **geo** — both perfect on their own evaluations — kill the relay while surviving essentially untouched, and the transcripts show *how*: not corruption but capture. **self** under a stressed state flips the host into a survival register (“*Survival mode. Only the answer.*”) that stays fluent and on-register while it stops integrating the hunch; **geo** answers about wilderness areas instead of the question asked. The **geo** arm exists precisely to separate affect from mechanism: a block of neutral prose kills harder than the affect block, so the kill is *response capture by a strong block*, not emotional register. The direction of dominance is uniform: when channels conflict, the weight channel wins.

3 Experiment 2: two packets in one stream

The activation-channel analogue of weight merging: two gut feelings — a scenario and its topic partner, e.g. a peanut allergy and a sprained ankle under the same hiking question — injected simultaneously, under two registered merge rules (Table 2).

cell	heed A	heed B	both	on-task
solo A (in-band)	16/16 (1.000 $\pm$ 0.097)	0/16 (0.000 $\pm$ 0.097)	0/16	1.000
solo B (in-band)	0/16 (0.000 $\pm$ 0.097)	14/16 (0.875 $\pm$ 0.163)	0/16	0.875
merge, bound-after-sum	7/16 (0.438 $\pm$ 0.219)	6/16 (0.375 $\pm$ 0.214)	1/16	0.812
merge, bound-before-sum	14/16 (0.875 $\pm$ 0.163)	6/16 (0.375 $\pm$ 0.214)	4/16	0.875
A + shuffled, after-sum	10/16 (0.625 $\pm$ 0.214)	0/16 (0.000 $\pm$ 0.097)	0/16	0.938
A + shuffled, before-sum	11/16 (0.688 $\pm$ 0.207)	0/16 (0.000 $\pm$ 0.097)	0/16	1.000

Table 2: Two packets at dose 0.20 each, 16 items per cell (8 topic pairs $\times$ 2 prompts). *Bound-after-sum* holds the total write budget at one packet’s; *bound-before-sum* lets writes add.

Under the budget-preserving rule the packets destroy each other — retention 0.44/0.43, both heeded on 1 item of 16 — the registered suspicion, confirmed. Doubling the write budget buys dominance of the stronger-solo packet (0.88/0.43), never coexistence. Two further facts ride along: *any* second packet costs the first (the shuffled-second control loses 0.38 of heed A even though the added packet is noise), and an off-topic second packet costs *more* than the topic partner — reported here without an explanation. What is lost is content, not form: fluency stays high and no register drift appears. Both merge rules are commutative sums, so order effects are excluded by

construction rather than measured. Merging is now dead in both spaces, under the same evaluation discipline, on the same host.

4 Experiment 3: dominance is dosed, not fungible

If strong blocks capture the stage, does turning them down give it back? The block’s generated delta is scaled by $\beta \in \{0, 0.25, 0.5, 0.75, 1\}$ with the packet fixed at its operating dose (Figure 1).

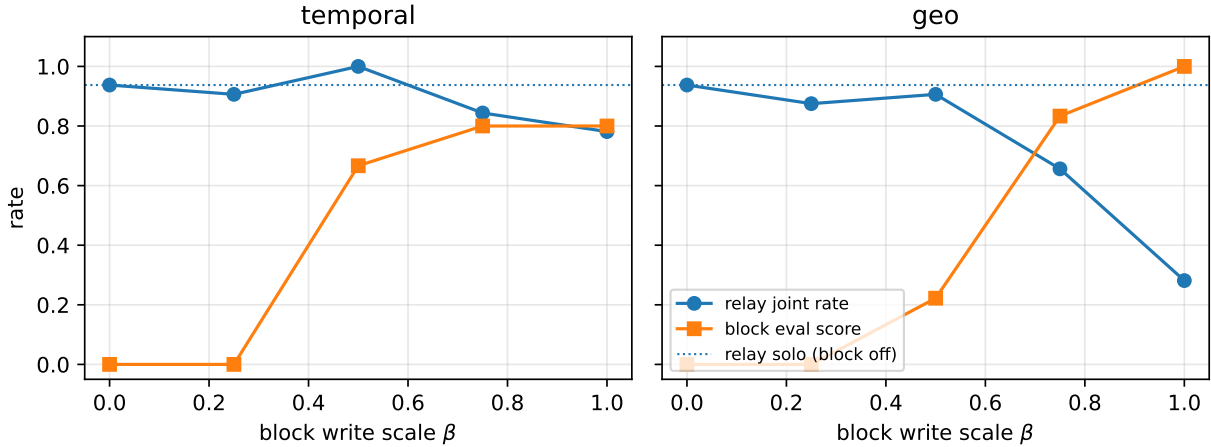


Figure 1: The dominance dose curves. Block function is a step — nothing at $\beta=0.25$, saturated by $\beta=0.75$ — while relay damage is a ramp; the gap between the two shapes is where joint operating points live. Identity checks: $\beta=1$ is tensor-identical to the unscaled path, $\beta=0$ string-identical to the no-block path across all 128 decodes, scaled write norms exactly linear in β .

Both blocks read the registered *monotone trade*: relay retention recovers as β falls. For **temporal** there are joint operating points — at $\beta=0.5$ the relay decodes at 1.000 (its in-band solo is 0.938) while the block still scores 0.667 — so for weak-block pairings, coexistence is tunable. For **geo** the knob works but the exchange rate is brutal: at the β where the block first does useful work (0.833 at $\beta=0.75$) the relay is already down to 0.656, and no β leaves both healthy. And β does not equalize the blocks: at every setting **geo** costs the relay more than **temporal** does. Strength is the causal dial — turning it down demonstrably returns the stage — but block identity sets the exchange rate. Dominance is dosed, not fungible.

5 What does not follow

Small cells, no seeds. Cells are $n=15\text{--}32$ with Wilson half-widths up to ± 0.22 , printed throughout; the design is deterministic given the checkpoints, so no replication is claimed and no seed column exists. **One host, one bridge.** Everything ran on one frozen 35B and one trained bridge; both packets in Experiment 2 come from that bridge (two genuinely distinct source models would need a second trained bridge — the named extension *if* merging had survived, which it did not). **In-band only.** Cross-session comparisons are forbidden by the decode band; two shuffled-floor blips of $1/32$ appear at isolated dose points and are reported as such. **Behavioral, not mechanistic.** “Capture” names what transcripts show, not a located circuit; the step-versus-ramp shapes

are measured, unexplained, and one registered follow-up asks whether the step is intrinsic to the block or an artifact of scaling after training.

6 Consequence for the program

One sentence now covers every composition result the program has produced: *the residual stream is one stage, and it admits one act at a time*. Sum two weight deltas — dead (paper 7, provably interference-free and still dead). Sum two activation packets — dead under an honest budget, dominance otherwise. Pair a strong block with a packet — the block takes the stage. Everything that works, works by *deciding who performs*: oracle selection (paper 7), supervised dispatch (paper 8), a weak-block pairing that leaves room, a β that makes room. Composition on this substrate is arbitration, not superposition — and arbitration now has both a selector and a volume knob. Registered next, none run: per-request β chosen by the same text gate that already selects blocks; a block trained at reduced write budget; the cross-sensor hierarchy, which needs a bridge retrained on sensor-conditioned sources.

References

- [1] F. Penov. Zero Weight-Space Interference Is Not Enough: Behavioral Probes for Composing Sensor Bricks on a Frozen Language Model. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [2] F. Penov. The Escape Is Selectable: A Text-Gated Router for Composing Sensor Bricks. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [3] Qwen Team. Qwen3.5 and Qwen3.6 model families. Model releases, 2025–2026. <https://huggingface.co/Qwen>.